Nivedha M

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Professional Summary

Electronics and Communication Engineering student with strong interest in VLSI design and digital systems. Skilled in RTL design using Verilog HDL, FSM modeling, and simulation. Seeking opportunities in semiconductor and chip design domains.

Education

B.E. Electronics and Communication Engineering 2023–2027
KSR Institute for Engineering and Technology, Tamil Nadu CGPA: 8.36

Higher Secondary Examination 2022–2023
Hillfort Matric Hr. Sec. School 81.33%

Technical Skills

- **VLSI:** RTL Design, Verilog HDL, Synthesis, Simulation, Functional Verification, Timing Analysis
- **Embedded Systems & IoT:** Embedded C, ESP32, Microcontrollers, Sensor Interfacing
- **Tools:** EDA Playground, MATLAB, Arduino IDE, Xilinx Vivado.

Project Experience

Smart Fire and Gas Detection System (ESP32, IoT)

- Designed IoT-based safety system using ESP32, MQ gas and flame sensors.
- Reduced detection latency by 30% through optimized Embedded C algorithm.
- Enabled real-time Wi-Fi alerts with response time under 3 seconds.

Smart Door Lock System (ESP8266)

- Developed ESP8266-based fingerprint smart door lock with 98%+ authentication accuracy and sub-500ms response time.
- Implemented IoT-based relay control with real-time biometric validation, eliminating unauthorized access during testing.

Internship Experience

VLSI Design Intern [Taras System and Solutions]

- Designed and simulated digital circuits using Verilog HDL, including combinational and sequential logic modules.
- Performed RTL design, functional verification, waveform analysis, and debugging using EDA tools.
- Worked on FSM-based designs and gained exposure to synthesis and digital design flow.
- Developed practical knowledge of front-end VLSI design from specification to verification.

VLSI Intern [Caliber Embedded Technologies]

- Developed and simulated 5+ RTL modules using Verilog HDL.
- Implemented combinational and sequential circuits with successful functional verification.

IoT Intern [Samcore Sollution]

- Integrated multiple sensors with microcontrollers and improved signal stability through debugging.

Certifications

- NPTEL – Internet of Things (Elite)
- Workshop on IoT and Embedded System
- EduSkills – Microchip

Relevant Course Work

• Digital Electronics – Logic gates, Flip-flops, Counters, Multiplexers, Decoders

• VLSI Design – RTL design flow, Synthesis, Timing Analysis, Physical Design basics